



Visual Design Process Transcript

Hello everyone, welcome to BITS PLANET DIGITAL. Today we will continue our learning journey for the course data visualisation and storytelling. We will be now looking at the module building dashboards and continue our concepts around building effective dashboards.

We will now look at the visual design process and illustration of a visual design process. Let me start with this. In our earlier classes, we have seen Edward Tufte's visual encoding principles and how to improvise the effectiveness of our visuals.

We have seen the strategic use of colour, size and shape and how different pre-attentive attributes can be used to enhance the appeal of our visuals. We will be now translating those principles for designing effective dashboards. However, there are certain extra elements that would come when we talk about dashboards and our focus will be the same in this class.

We will talk about design goals for a dashboard. We will be now looking at design goals for making a dashboard. The primary design goal for any dashboard is driving insight and action from the data.

In this regard, we will be borrowing a key principle from Edward Tufte who says that we should reduce or improvise the information content in the visual by removing the redundant information. And he has proposed a principle to identify and reduce this redundant or excessively useless information on the graphic. It is called data to ink ratio principle.

What does it mean? It says that every additional ink that you use for printing your graphic should go towards the visualisation or a particular feature of the visualisation that enhances the information in the visualisation. If it is not enhancing, you should not give that pre-attentive attribute. For example, use a darker shade or use a vibrant hue of colours when the colours don't mean anything.

And you can print it in simple one colour or for very best in simple grey colour so that the contents are effectively reflecting the insights without either creating overemphasis or lower emphasis. What relates to ink as for graphics in the context of dashboard you can think of as pixels. Why? Because you would be operating on a digital screen where the pixel density or the pixel space determines the effectiveness.

The pre-attentive attributes that you use for a dashboard are determined by the pixels that you use for colouring, bringing in contrast or showing the depth of the object. All these things would require higher pixels. So, use lesser ink for a visual, use lesser pixels for a dashboard.

Only you should use them in addition if they are bringing more information to the dashboard or the visual. Having seen the design principles, now we will see how to translate them as design goals and implement them. When we say the data ink on a graphic is equal to data pixel on a visualisation, we have to rationalise the non-data pixels by which you have to remove the attributes or modify the visualisation parts of visualisation which are not enhancing any data inside and reduce the pixel density there or use lighter colours, simple borders, enhance white space so that they get de-emphasised.

Use the opposite to enhance areas where there is data, insight, highlight that you want to give. We will now look at this through an example which is given on the slide. We will also test this in our experiential learning sessions so that you know how to best use the data pixel ratio to improve the effectiveness of your dashboard.

So, this particular dashboard is taken from Corda Airlines dashboard and this example is from Stephen Hugh information dashboard design textbook. So, this graphic you can very well understand has some key elements but it is visually not that pleasing or has strikingly different colours and it has very cluttered images. The user is not able to readily know where to go and look out for what information.

We will now identify the non-value adding elements in this particular dashboard. We will look at any particular visual attribute or a graphical design or the presentation of a graphic that can be read, de-emphasised without losing the information content or enhancing a particular information that we want to highlight. First and foremost, there is no need for a 3D diagram in a 2D space.

This creates unnecessary disorientation to the viewer by removing the 3Ds on graphs and pie. We do not use in information, we do not lose any information but in fact we would be enhancing the visual appeal of the graphic. Similarly, the grid line in the bars is irritating and does not particularly add value.

This being a dashboard, you can add the pop-up or the queue when somebody moves a cursor over there and even in fact remove the axis or provide more context to what this means to the traveller or to the airline. Finally, there is no need for decoration in the background with this kind of rectangular blocks showing different colour and not adding any particular context or value to that particular graph. And the colour gradients in the background are also displeasing and distorting the view.

A plain background would have done a much more better job and give more appeal to the graphics that are conveying the information, highlighting them. Contrast does not mean you use only colour. A plain background gives the best contrast when you want to highlight a particular area in the graph.

And the background finally does not enhance the aesthetics. So, one has to relook at this such kind of dashboard designs from a aesthetic standpoint and see the overall graphic is pleasing to the user. So, now we will talk about use of visual attributes to enhance the designs within the dashboard.

Whenever you are talking about a dashboard space, you can use contrast for 3-4 things. First is to emphasise a particular content as opposed to others. Then you can also do a de-emphasise by completely moving it into the background or reducing the saturation, colour saturation, so that naturally we don't perceive as audience on that particular region.

Or you can have a neutral area which is neither emphasised or de-emphasised by using a mix of saturations. The graphic on to your right shows how to create emphasis. And in fact, you can create much more particular emphasis by creating a border around the graphic and highlighting it in white colour, visually drawing attention of the audience to that particular region.

These are different types of uses of contrast and colour to bring emphasis on a sub-regions of the dashboard where you want to draw attention or you want to de-emphasise that region so that the attention is not spent on that particular area. Typical rule of thumb or guiding principles for bringing more emphasis on to the dashboard, always use the top left centre areas of the dashboard to receive the greatest emphasis. Top left and centre areas of the dashboard should receive the greatest emphasis with key information going into those areas.

Because if you remember cognitive abilities of humans and types of memory, we are attuned to read the information from top left to right and top to bottom when we look at any information. It is the preconditioning of the human brain that makes you to read any information from top left to right and from top to bottom. So by arranging the key elements that you want to highlight on your dashboard at the top left corner, you are bringing natural attention of your audience to that particular region.

Similarly, you can also draw central attention to a central region by creating a border around it. So central regions are also a very attentive spaces within the dashboard, but you need to create a border around it to bring much more emphasis. And avoid placing non-essential items like logos and navigational controls in the prime areas because this is where the natural attention would go.

So, you should give the most highlighted information in that space and remove the other important elements from those areas so that the overall effectiveness of your visual improves. Finally, we have seen the use of various visual attributes to enhance the effectiveness of visuals. These are taken from your pre-attentive attributes lesson.

So, one can use orientation, a slight deviation in the orientation will naturally bring the attention, change in shape, change in the length of the line, thickness of the line, size of the dot or the size of the shape, curvature, added marks, enclosure, hue, intensity, spatial position and motion. All these elements can be used on the dashboard or the graphics within the dashboard, then you can create highlighted spaces within your dashboard to bring focused attention or to de-emphasise a particular region right or create contrast between different types of graphs that are on the dashboard to bring attention of your audience. A couple of examples colour intensity generally higher colour intensity is pursued for a strongest feature.

For example, it would be ideal to indicate the higher growth rate regions with dark green colour than with light green colour naturally indicating higher growth with green colour dark green colour. Similarly, size bigger is the size of certain things on dashboard higher is the importance automatically assigned to it. For example, you can highlight the pollution or the CO2 emissions with larger size of bubbles as compared to a company or a fact or a country emitting lower CO2 with a smaller bubble size.

Similarly, line width you can talk about the current year funding or the current year target with a thicker line and the current actual growth with a lighter line to show that what is the actual gap or how is the performance changing over a period of time. One has to strategically use these different types of attributes to bring the attention on a particular graphic for the audience. So, that the insights that you want them to pick from the data are intuitively picked by the audience.

We will now see the application of these principles on a particular dashboard which shows the changes in life expectancy across OECD countries at birth at the age of 40 at the age of 60. And I have chosen to show the difference between life expectancy at the age of 60 categorised by gender men and women. So, you can see there are two categories the quantitative variable is the life expectancy the main categories of men and women and the layering category or the main superimposing category is the country type which is the OECD country.

Then also you have the time as a category over a period of time how is it moving. So, you can see clearly over a period of time in 2000 the life expectancy by ah at the age of 60 for women has been increasing. So, has been for men.

However, women are higher than men clear visualisation clearly showing the attributes that women are having higher life expectancy than men and it is valid across time periods. Similarly, you can see the trend across the ah countries and if you highlight a particular country you can see the number what is the actual ah number of years by which the men ah women are expected to live. So, than the men which is indicated by the graphic at the lower and the OECD average is indicated in a different colour clearly distinguishing the countries below the average and the above the average.

So, all these attributes if you see the colour size shape have been used. So, colour has been used size and shape to distinguish between men and women then there is a contrasting feature for the OECD average and you have the interactivity and the clear choose of text with highlighted text showing the main variable and the description below to add more context. So, that the reader is able to readily get the information from the graphic.

So, simple presentation should be the goal of your dashboard display and it tries to always display large amount of data in a small space. So, you have to be very clear about how and what to present and where to present. So, in this case just by highlighting life expectancy by age and gender when you look at this graphic your attention rightly goes to this particular top left corner and you know this is the variable that I am looking at.

And, then you have the characteristics you have the cue how to read the graph men are rhombus in shape women are circle dot and they are differently coloured. So, it is clearly effective and expressive in the sense there could have been better ways to do this, but in the given time and space this is the most effective way to present the visual and it is capturing all the information that you want to convey. You want to convey the changes over a time and changes across the country and showing how the average is moving.

And, it is the best way to display information that is found in the dashboards and it serves the purpose even when it is fitted into as a small space. So, with this you can try to improve the effectiveness and expressiveness of your dashboard.

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